

Moment-to-Moment Affect During Film-Induced Somatic Anxiety

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Abstract

Studying sustained anxiety requires tracing how unpleasant arousal develops, persists and subsides over an unfolding experience. We characterised a 13.5-min first-person film of people undertaking dangerous activities at height as an induction for this purpose. Using an open-source browser-based platform developed for this study, online participants continuously rated their own valence and arousal and completed the somatic subscale of the State-Trait Inventory for Cognitive and Somatic Anxiety before and after viewing. In the confirmatory sample ($N = 124$), arousal was elevated during the film's rising and sustained phases and declined towards the end, while valence was more unpleasant during the sustained phase than during the opening. Group-mean trajectories closely reproduced the temporal shape observed in an exploratory sample ($n = 21$; arousal $r = .99$; valence $r = .94$). Somatic anxiety increased after viewing ($d_z = 1.03$). Participants reporting larger increases rated the film as more arousing ($\rho = .24$) and more unpleasant ($\rho = -.38$). The open-source platform and reference trajectories provide a reusable resource for studying the temporal course of affect during a sustained anxiety induction.

Keywords: emotion induction, film, continuous ratings, valence and arousal, somatic anxiety

Introduction

Studying sustained anxiety requires an account of how the response develops and subsides during an unfolding experience. Its timing and duration matter when relating felt emotion to concurrent behaviour or brain activity (Saarimäki, 2021). Understanding how bodily signals contribute to such experience likewise requires inducing arousal that changes over time and measuring it under ecologically valid, naturalistic conditions (Cristaldi et al., 2026). Films offer a repeatable sequence of events through which to study these changes. Established film libraries demonstrate their value for eliciting emotion (Gilman et al., 2017; Gross & Levenson, 1995; Schaefer et al., 2010). Characterising the response over time adds to that evidence by locating periods of sustained unpleasant arousal within a particular film.

Continuous self-report is an established way to trace affect during viewing (Ruef & Levenson, 2007), and has been used to relate emotional experience to neural responses (Goldin et al., 2005). Two-dimensional ratings allow viewers to report pleasantness and activation together, using a grid grounded in the valence–arousal framework (Russell, 1980; Russell et al., 1989). Continuous measurement in this space has a history in music research (Schubert, 1999) and media annotation (Girard & Wright, 2018). For a film to serve as a reference stimulus in studies of sustained anxiety, it is useful to establish whether different groups show a similar temporal response and whether that response accompanies anxiety symptoms.

The present film shows people climbing and performing unsecured stunts on a skyscraper. Viewers remain physically safe while watching others in danger, a situation that may evoke anticipation, concern for the people depicted and bodily discomfort. To characterise this response, we measured viewers' valence and arousal continuously and assessed somatic anxiety before and after viewing with the State-Trait Inventory for Cognitive and Somatic Anxiety (Grös et al., 2007; Ree et al., 2008). This combination addresses both the unfolding affective response and the change in reported bodily symptoms, including a fast heartbeat and muscle tension. A retrospective rating of anxiety during viewing provides a further link to the intended induction.

Here, we asked whether a sustained heights film elicits a reproducible course of unpleasant

arousal accompanied by increased somatic anxiety. We developed an open-source browser-based platform for continuous valence and arousal ratings, using an exploratory sample to inform hypotheses tested in an independent confirmatory sample. We predicted that arousal would rise, remain elevated and then decline, that the sustained period would be more unpleasant than the opening, and that somatic anxiety would increase after viewing. We also expected the temporal pattern of affect to reproduce across samples. To connect these dynamics to individual experience, we predicted that everyday fear of heights and increases in somatic anxiety would be associated with higher arousal and more unpleasant valence, and that retrospectively reported anxiety would converge with heightened arousal and somatic change. Together, these tests assess the film's suitability for studying sustained anxiety and the relationship between its affective and somatic components.

Materials and Methods

Preregistration and Transparency

The hypotheses and analysis plan were preregistered on OSF (<https://osf.io/s2yvb>). All confirmatory tests followed the registered analysis plan; exploratory analyses are identified as such.

Participants

Participants were recruited on Prolific (Palan & Schitter, 2018), with filters for fluent English, normal or corrected-to-normal vision, an approval rate of at least 95%, at least 20 previous submissions and a desktop computer with a mouse or trackpad. They were recruited in cohorts with a 50/50 quota on Prolific's sex screener and were paid £4.80. The information sheet advised people with a strong fear of heights not to take part. The pilot and the first cohort formed the exploratory sample, which was used to fix the segment edges, exclusion rules and sample size; later cohorts formed the confirmatory sample, on which all confirmatory tests were run. Recruitment stopped at the end of the first cohort after which at least 120 usable confirmatory participants were available, a number giving 80% power to detect correlations of about .25. The study involved watching a film and answering questionnaires and was not a health research project

under the Danish Act on Research Ethics Review of Health Research Projects and Health Data Research Projects; for such studies there is no legal requirement for ethical approval in Denmark, and Aarhus University does not require it. The Aarhus University Research Ethics Committee provides a standard statement to this effect. All participants gave informed consent online before taking part.

Stimulus

The stimulus was a 13 min 28 s first-person video of an unsecured climb to the roof of a skyscraper (Tikhomirov, 2017), presented without sound (1280×720 pixels, 25 frames per second) and downloaded in full before playback.

Procedure and Measures

The task ran in an ordinary web browser; it was written for this study with the open-source jsPsych library (de Leeuw et al., 2023) and hosted on a JATOS server (Lange et al., 2015). After consent and a device check, participants rated their everyday fear of heights ('In everyday life, how afraid are you of heights?', 1–7) and practised the rating task (tracking a moving target and placing example states on the grid). They then completed the somatic subscale of the State-Trait Inventory for Cognitive and Somatic Anxiety (STICSA; 11 items, 1–4, total 11–44, state instructions; Ree et al., 2008), watched the film while rating continuously, and completed the STICSA again immediately afterwards. During the film, participants moved a dot within a small affect grid (Russell et al., 1989) in the corner of the video (pointer locked; Supplementary Figure S1); horizontal position coded valence (0 = unpleasant, 100 = pleasant) and vertical position arousal (0 = low, 100 = high). Instructions asked participants how they felt at that moment. Position was sampled at 20 Hz against film time; leaving full screen or switching tabs paused the film. Afterwards, participants rated how tense or anxious they had felt overall while watching (1–7) and answered further questions on their experience and data quality. Participants also provided a written account of what they saw and how the film made them feel (minimum 100 words), with a separate optional field for feedback on the study or technical difficulties. Analyses of the narrative content are outside the scope of the present report.

Data Processing and Exclusions

Ratings were resampled to 1 Hz of film time (808 values) by linear interpolation. The film was divided into four preregistered segments, estimated in the exploratory sample by piecewise-linear segmentation of group-mean arousal: baseline (0–262 s), rise (262–470 s), plateau (470–692 s) and return (692–808 s). These are descriptive labels; baseline denotes the film’s opening. Participants were excluded if they failed both attention checks, produced script-generated input, did not type their written description (fewer than 0.2 keystrokes per character for descriptions of at least 200 characters), gave erratic input (more than 10% of mouse movements larger than 250 pixels), had disrupted viewing (more than 30 s of buffering or 60 s of interruptions), had ratings during fewer than 95% of the film, or had seen the film before.

Statistical Analysis

All tests were two-sided ($\alpha = .05$) with Holm correction within each hypothesis family; directional hypotheses counted as supported only if significant in the predicted direction. H1 compared STICSA totals after and before the film (paired t test; d_z with exact 95% CI; Wilcoxon signed-rank test as a check). H2 tested four contrasts between participants’ segment means (arousal: rise > baseline, plateau > baseline, return < plateau; valence: plateau < baseline) with paired t tests. H3 correlated the confirmatory and exploratory group-mean series (Pearson r ; 95% CI from 5,000 circular block bootstrap resamples with 30-s blocks) and was supported if the lower bound exceeded .50. H4–H6 were Spearman correlations with bootstrap 95% CIs (5,000 resamples): everyday fear of heights with plateau arousal, plateau valence and STICSA change (H4); STICSA change with whole-film arousal and valence (H5); and post-film anxiety with STICSA change and plateau arousal (H6). Two registered sensitivity samples (stricter engagement criteria; participants rejected on Prolific removed) repeated all tests. Exploratory analyses examined the change in each STICSA item (Wilcoxon signed-rank tests, Holm-corrected across items) and the effects of age and sex (mixed models of segment means). Analyses were run in Python 3.13.7 with NumPy, SciPy, pandas and statsmodels.

Results

Of 130 confirmatory participants who completed the study, 6 were excluded (3 disrupted viewing, 2 untyped description, 1 erratic input), leaving $N = 124$ (mean age 34.2 years, $SD = 12.0$; 59 female, 62 male, 3 not available). Sample characteristics, descriptive statistics and all confirmatory tests are given in Supplementary Tables S1–S3.

Time course (H2, H3). Arousal was higher in the rise segment than in the opening segment (labelled baseline in the preregistration), $t(123) = 14.67$, $p_{\text{Holm}} < .001$, $d_z = 1.32$, 95% CI [1.07, 1.56], and in the plateau than in the opening, $t(123) = 19.77$, $p_{\text{Holm}} < .001$, $d_z = 1.78$, 95% CI [1.49, 2.06]. It was lower in the final segment than on the plateau, $t(123) = -14.70$, $p_{\text{Holm}} < .001$, $d_z = -1.32$, 95% CI [-1.56, -1.08] (Figure 1A). Valence was lower on the plateau than in the opening segment, $t(123) = -9.85$, $p_{\text{Holm}} < .001$, $d_z = -0.88$, 95% CI [-1.09, -0.68] (Figure 1B). Supplementary Figure S4 shows, as an exploratory description, where on the affect grid participants' ratings fell in each segment. The group-mean series of the confirmatory sample closely followed the shape of the exploratory sample's series, for arousal, $r = .993$, 95% CI [.990, .996], and for valence, $r = .937$, 95% CI [.893, .966]. Split-half reliability of the group-mean series was .992 for arousal and .979 for valence.

Somatic anxiety (H1). STICSA somatic scores rose by 6.79 points from before ($M = 13.76$, $SD = 3.56$) to after the film ($M = 20.55$, $SD = 7.29$), $t(123) = 11.46$, $p < .001$, $d_z = 1.03$, 95% CI [0.81, 1.25] (Figure 2A). Scores rose for 109 participants and fell for 8, and the Wilcoxon test agreed ($T = 197$, $p < .001$). In an exploratory item analysis, all eleven items increased ($p_{\text{Holm}} < .001$; Figure 2B and Supplementary Table S6), most of all a fast heartbeat ($d_z = 1.17$) and least a dry throat, weak muscles and stiff limbs ($d_z = 0.52$ – 0.54).

Individual differences (H4–H6). Participants whose STICSA scores rose more gave higher whole-film arousal ratings, $\rho = .24$, 95% CI [.05, .41], $p = .008$, $p_{\text{Holm}} = .008$, and lower valence ratings, $\rho = -.38$, 95% CI [-.51, -.23], $p < .001$, $p_{\text{Holm}} < .001$. Post-film anxiety correlated with STICSA change, $\rho = .56$, 95% CI [.44, .67], $p < .001$, $p_{\text{Holm}} < .001$, and with plateau arousal, $\rho = .34$, 95% CI [.15, .50], $p < .001$, $p_{\text{Holm}} < .001$. Everyday fear of heights was

associated with more negative plateau valence, $\rho = -.32$, 95% CI $[-.46, -.16]$, $p < .001$, $p_{\text{Holm}} = .001$; its associations with plateau arousal, $\rho = .08$, 95% CI $[-.09, .26]$, $p = .355$, $p_{\text{Holm}} = .355$, and with STICSA change, $\rho = .18$, 95% CI $[-.00, .35]$, $p = .049$, $p_{\text{Holm}} = .098$, were not significant after correction.

Sensitivity and additional analyses. All registered support decisions were unchanged in both sensitivity samples ($n = 96$ and 120 ; Supplementary Table S4). In a registered descriptive analysis, re-estimating the segment edges in the confirmatory sample placed them at 262, 467 and 660 s, against the registered 262, 470 and 692 s. Exploratory analyses found no evidence that the time course differed by sex (arousal: $\chi^2(3) = 0.28$, $p = .964$; valence: $\chi^2(3) = 0.72$, $p = .869$) or by age, or that STICSA change was associated with sex or age (Supplementary Table S5 and Supplementary Figures).

Discussion

Watching people undertake dangerous activities at height elicited sustained unpleasant arousal and a substantial increase in self-reported somatic anxiety. Across two independent samples, viewers' affect followed a closely similar course: arousal rose, remained elevated for several minutes and then declined, while valence was more unpleasant during the sustained period than during the opening. Viewers who experienced the film as more unpleasant and arousing also reported larger increases in bodily symptoms. These findings build on established uses of film to induce emotion (Gross & Levenson, 1995; Schaefer et al., 2010), showing that a sustained naturalistic experience can provide a reproducible temporal structure for studying anxiety. Because this structure is tied to reported bodily symptoms, the film and the open-source platform used to characterise it offer a way to ask how physiological changes and their neural representation relate to an unfolding emotional experience.

Somatic anxiety increased substantially after viewing ($d_z = 1.03$), with increases across all eleven symptoms in an exploratory item analysis and the largest change in reports of a fast heartbeat. Participants with larger somatic increases reported higher average arousal and more unpleasant valence during the film, and retrospective anxiety ratings converged with both somatic

change and plateau arousal. Everyday fear of heights was associated with more unpleasant plateau valence, although its other associations did not meet the corrected threshold. Anticipation, concern for the climbers and discomfort elicited by views of height may all contribute to the experience.

The combination of sustained unpleasant arousal and increased somatic symptoms makes this film a promising induction for interoception research, which examines how signals from the body and their perception contribute to emotional experience (Critchley & Garfinkel, 2017). Recent work has called for inductions that produce dynamic changes in arousal, together with ecological measures of how people experience their bodily signals (Cristaldi et al., 2026); the present film provides such a sequence. Recording cardiac, respiratory and electrodermal activity during viewing (Kreibig, 2010) would allow future studies to examine how physiological responses relate to the shared affective trajectory and to individual differences in reported bodily experience. Combined with fMRI, this approach could test how relationships between bodily signals, brain activity and subjective affect change as arousal builds, persists and subsides. Such work would extend established approaches linking continuous emotion ratings to neural responses during naturalistic viewing (Goldin et al., 2005; Saarimäki, 2021).

The open-source platform supports these applications by collecting simultaneous valence and arousal ratings within the video, synchronised to film time. It complements desktop media-annotation tools (Girard, 2014; Girard & Wright, 2018) and immersive virtual-reality tools (Fourcade et al., 2025), enabling online characterisation of stimuli before their use in laboratory studies. The reference trajectories can guide the selection of film intervals or provide affective regressors for independent neuroimaging samples, alongside relevant visual and event-related features. The platform can also be used with other films to examine which aspects of the temporal response generalise across stimuli.

Several limitations remain. Without a control film, we cannot isolate the contribution of heights or establish specificity relative to other intense films. The effects of continuous rating should also be evaluated before transferring these trajectories to passive viewing (Hutcherson

et al., 2005). Somatic symptoms were assessed only before and after the film, leaving their temporal course and relationship to physiology unresolved; 36% of participants also began at the scale minimum and could not report a decrease. Group-mean trajectory correlations establish similarity of temporal shape rather than absolute agreement or individual reliability; the confirmatory group, for example, reported somewhat higher arousal during the opening and final minutes (Figure 1). The final decline also does not establish return to baseline. Generalisation is limited by online recruitment and the advice that people with a strong fear of heights should not participate.

The film and platform provide a reusable approach to studying sustained unpleasant arousal alongside changes in somatic anxiety. Their value for future work lies in bringing a reproducible emotional time course into studies of how bodily responses and brain activity contribute to subjective experience.

Disclosure of Interest

The author reports there are no competing interests to declare.

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Generative AI Declaration

Generative AI tools (ChatGPT and Codex, OpenAI; Claude Code, Anthropic) assisted with study planning, development of the online rating platform and analysis code, data analysis and visualisation, literature searches, and manuscript preparation. The author reviewed and edited the AI-assisted materials, verified the analyses and references, and takes full responsibility for the accuracy and integrity of the work.

Data Availability Statement

The preregistration is available at <https://osf.io/s2yvb>. The platform code is available at <https://github.com/emodied-computation-group/online-affect-rating> (version

v1.0-heights-film). The analysis code and anonymised participant data are available at <https://github.com/embodied-computation-group/heights-film-paper> and archived on Zenodo (<https://doi.org/10.5281/zenodo.23151387>). The film is a third-party video that is publicly available on YouTube (Tikhomirov, 2017) and is not redistributed.

Author Contributions (CRediT)

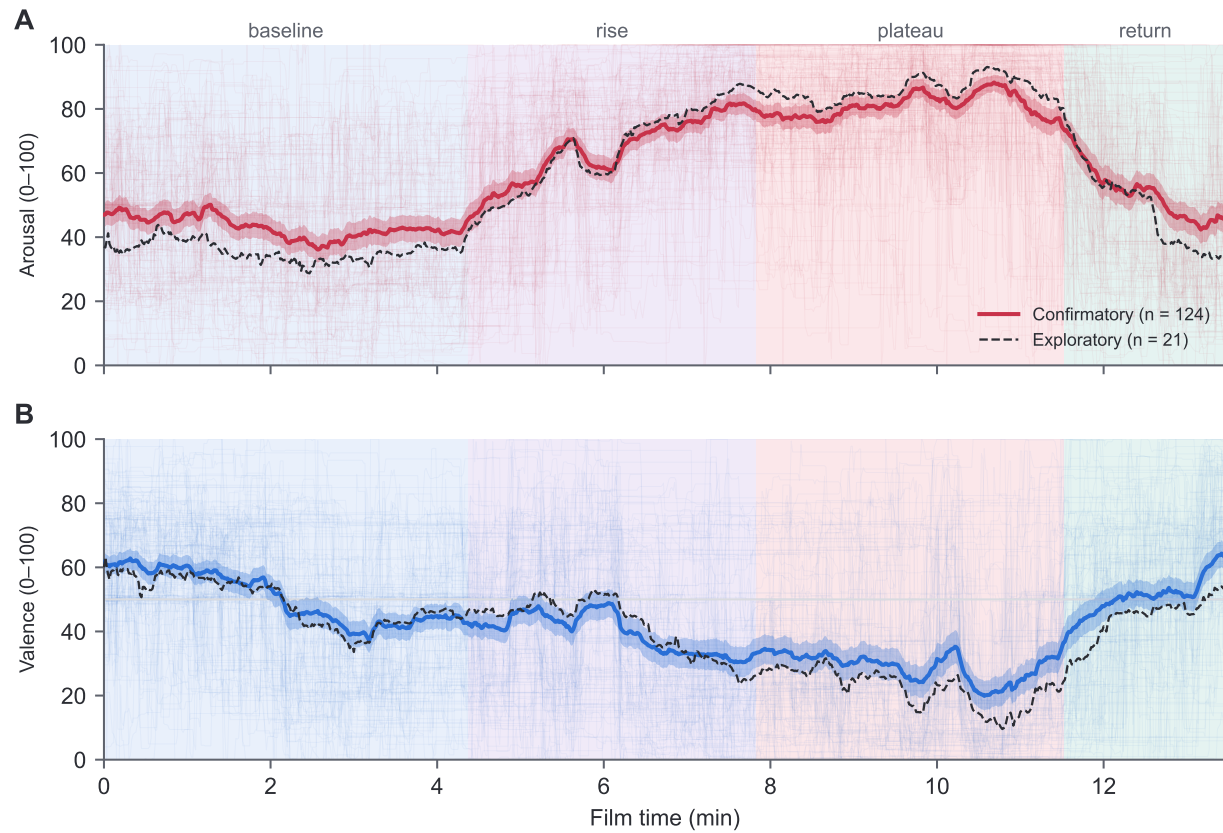
Micah G. Allen: Conceptualisation, Methodology, Software, Investigation, Data curation, Formal analysis, Visualisation, Writing – original draft, Writing – review and editing, Project administration, Funding acquisition.

References

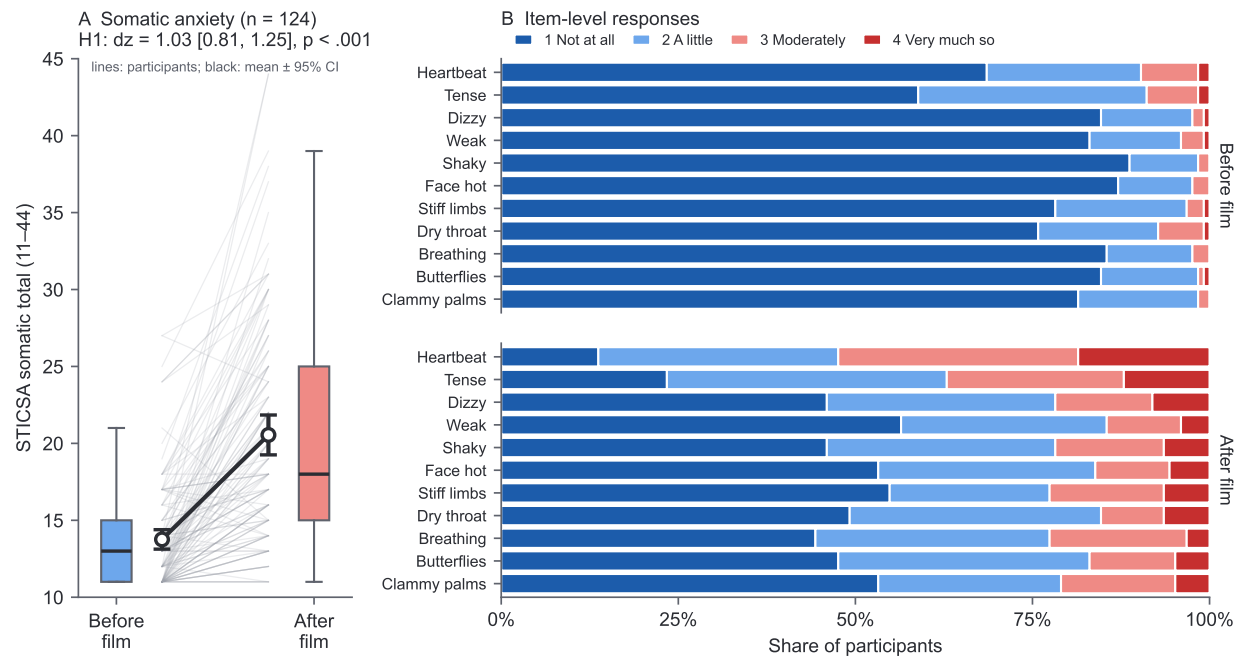
- Cristaldi, F. D. P., Banellis, L., Bottesi, G., & Allen, M. G. (2026). Beyond accuracy: Interoceptive self-appraisal and the meaning of bodily signals. *Trends in Cognitive Sciences*. <https://doi.org/10.1016/j.tics.2026.09.010>
- Critchley, H. D., & Garfinkel, S. N. (2017). Interoception and emotion. *Current Opinion in Psychology*, 17, 7–14. <https://doi.org/10.1016/j.copsyc.2017.04.020>
- de Leeuw, J. R., Gilbert, R. A., & Luchterhandt, B. (2023). jsPsych: Enabling an open-source collaborative ecosystem of behavioral experiments. *Journal of Open Source Software*, 8(85), 5351. <https://doi.org/10.21105/joss.05351>
- Fourcade, A., Malandrone, F., Roellecke, L., Ciston, A., de Mooij, J., Villringer, A., Carletto, S., & Gaebler, M. (2025). AffectTracker: Real-time continuous rating of affective experience in immersive virtual reality. *Frontiers in Virtual Reality*, 6, 1567854. <https://doi.org/10.3389/frvir.2025.1567854>
- Gilman, T. L., Shaheen, R., Nylocks, K. M., Halachoff, D., Chapman, J., Flynn, J. J., Matt, L. M., & Coifman, K. G. (2017). A film set for the elicitation of emotion in research: A comprehensive catalog derived from four decades of investigation. *Behavior Research Methods*, 49(6), 2061–2082. <https://doi.org/10.3758/s13428-016-0842-x>
- Girard, J. M. (2014). CARMA: Software for continuous affect rating and media annotation. *Journal of Open Research Software*, 2(1), e5. <https://doi.org/10.5334/jors.ar>
- Girard, J. M., & Wright, A. G. C. (2018). DARMA: Software for dual axis rating and media annotation. *Behavior Research Methods*, 50(3), 902–909. <https://doi.org/10.3758/s13428-017-0915-5>
- Goldin, P. R., Hutcherson, C. A. C., Ochsner, K. N., Glover, G. H., Gabrieli, J. D. E., & Gross, J. J. (2005). The neural bases of amusement and sadness: A comparison of block contrast and subject-specific emotion intensity regression approaches. *NeuroImage*, 27(1), 26–36. <https://doi.org/10.1016/j.neuroimage.2005.03.018>

- Grös, D. F., Antony, M. M., Simms, L. J., & McCabe, R. E. (2007). Psychometric properties of the State-Trait Inventory for Cognitive and Somatic Anxiety (STICSA): Comparison to the State-Trait Anxiety Inventory (STAI). *Psychological Assessment*, 19(4), 369–381.
<https://doi.org/10.1037/1040-3590.19.4.369>
- Gross, J. J., & Levenson, R. W. (1995). Emotion elicitation using films. *Cognition & Emotion*, 9(1), 87–108. <https://doi.org/10.1080/02699939508408966>
- Hutcherson, C. A., Goldin, P. R., Ochsner, K. N., Gabrieli, J. D., Barrett, L. F., & Gross, J. J. (2005). Attention and emotion: Does rating emotion alter neural responses to amusing and sad films? *NeuroImage*, 27(3), 656–668. <https://doi.org/10.1016/j.neuroimage.2005.04.028>
- Kreibig, S. D. (2010). Autonomic nervous system activity in emotion: A review. *Biological Psychology*, 84(3), 394–421. <https://doi.org/10.1016/j.biopsycho.2010.03.010>
- Lange, K., Kühn, S., & Filevich, E. (2015). “Just Another Tool for Online Studies” (JATOS): An easy solution for setup and management of web servers supporting online studies. *PLOS ONE*, 10(6), e0130834. <https://doi.org/10.1371/journal.pone.0130834>
- Palan, S., & Schitter, C. (2018). Prolific.ac—a subject pool for online experiments. *Journal of Behavioral and Experimental Finance*, 17, 22–27.
<https://doi.org/10.1016/j.jbef.2017.12.004>
- Ree, M. J., French, D., MacLeod, C., & Locke, V. (2008). Distinguishing cognitive and somatic dimensions of state and trait anxiety: Development and validation of the State-Trait Inventory for Cognitive and Somatic Anxiety (STICSA). *Behavioural and Cognitive Psychotherapy*, 36(3), 313–332. <https://doi.org/10.1017/s1352465808004232>
- Ruef, A. M., & Levenson, R. W. (2007). Continuous measurement of emotion: The affect rating dial. In J. A. Coan & J. J. B. Allen (Eds.), *Handbook of emotion elicitation and assessment* (pp. 286–297). Oxford University Press.
<https://doi.org/10.1093/oso/9780195169157.003.0018>
- Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178. <https://doi.org/10.1037/h0077714>

- Russell, J. A., Weiss, A., & Mendelsohn, G. A. (1989). Affect grid: A single-item scale of pleasure and arousal. *Journal of Personality and Social Psychology*, 57(3), 493–502.
<https://doi.org/10.1037/0022-3514.57.3.493>
- Saarimäki, H. (2021). Naturalistic stimuli in affective neuroimaging: A review. *Frontiers in Human Neuroscience*, 15, 675068. <https://doi.org/10.3389/fnhum.2021.675068>
- Schaefer, A., Nils, F., Sanchez, X., & Philippot, P. (2010). Assessing the effectiveness of a large database of emotion-eliciting films: A new tool for emotion researchers. *Cognition & Emotion*, 24(7), 1153–1172. <https://doi.org/10.1080/02699930903274322>
- Schubert, E. (1999). Measuring emotion continuously: Validity and reliability of the two-dimensional emotion-space. *Australian Journal of Psychology*, 51(3), 154–165.
<https://doi.org/10.1080/00049539908255353>
- Tikhomirov, A. (2017, February 22). *Most crazy girl ever Viki Odintcova. Dubai #6*. YouTube.
<https://www.youtube.com/watch?v=PDVsGyhteEg>

Figure 1*Continuous Arousal and Valence Ratings Over the Film*

Note. (A) Arousal (0 = low, 100 = high) and (B) valence (0 = unpleasant, 50 = neutral, 100 = pleasant), rated continuously and resampled to 1 Hz. Thin lines are individual participants of the confirmatory sample; the thick line is the group mean with its 95% confidence interval ($N = 124$); the dashed line is the group mean of the exploratory sample ($n = 21$). Shading marks the four preregistered segments, estimated from group-mean arousal in the exploratory sample. The segment names are descriptive labels; the opening segment is called baseline in the preregistration.

Figure 2*Somatic Anxiety Before and After the Film*

Note. (A) STICSA somatic totals (11–44) before and after the film in the confirmatory sample ($N = 124$). Grey lines connect each participant's scores; boxes show medians and interquartile ranges; the black line shows means with 95% confidence intervals. (B) Share of participants giving each answer (1 = not at all to 4 = very much so) to the eleven somatic items before (top) and after (bottom) the film; items are ordered as in the questionnaire.